Paper #26b: Sterile Neutrino Mass Generation â UQFF (Redirect Notice)

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Session: Phase 1 (Sessions 1â43)
Framework: UQFF Star-Magic (Îº = 0.0005/day, [SSq] = 0.57, Î²_i = 0.61)
Source: `CondensedPhysics2.py`, `MAIN_1_CoAnQi.cpp`
Cross-links: PAPER_026_Sterile_Neutrino_Mass_Generation_UQFF.md (canonical), PAPER_027 (Lepton Flavor Violation), PAPER_028 (BSM Coupling Constants)

Abstract

This file is a redirect stub. All sterile neutrino mass generation content â including the GUT-scale sector, Yukawa coupling matrix, seesaw mechanism, neutrinoless double beta decay constraints, and UQFF vacuum-mediated mass generation equations â has been merged into the canonical whitepaper PAPER_026_Sterile_Neutrino_Mass_Generation_UQFF.md. See that file for full derivations and numerical results.

Key parameters consolidated: sterile neutrino mass M_N ~ 1e14 GeV, active-sterile mixing |U_eN|Â² < 1e-3, vacuum contribution Î´m_Î½ = Îº Ã [SSq] Ã m_N.

> **MERGED INTO:** PAPER_026_Sterile_Neutrino_Mass_Generation_UQFF.md

Key Equations (Summary)

The UQFF-modified seesaw mass formula:

$$m_\nu = m_D^2/M_N + \delta m_{UQFF}$$

$$\delta m_{UQFF} = \kappa \times [SSq] \times \frac{v^2}{M_N} = 5.0 \times 10^{-4} \times 0.57 \times \frac{v^2}{M_N}$$

$$|U_{eN}|^2 < 1.0 \times 10^{-3} \quad \text{(experimental constraint)}$$

Numerical summary: M_N = 1.0e14 GeV, m_D = 1.74e2 GeV (top-quark scale), Î´m_UQFF â 2.85e-4 Ã vÂ²/M_N, Îº[SSq] = 2.85e-4

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UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?¹, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments. LHC direct-search results and arXiv constraints bound the parameter space.

Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
Îº	5.0 Ã 10â»â´ dayâ»Â¹	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
Î²_i	0.60â0.61	Buoyancy coupling coefficient
kâ	1.5	Ug1 DPM-dipole coupling
kâ	1.2	Ug2 outer-bubble charge coupling
kâ	1.8	Ug3 string-rotation coupling
kâ	2.0	Ug4 vacuum-concentration coupling
Î·	10â»Â²Â²	Inertia tensor scale
E_react(0)	10â´â¶ J	Reference reactive energy

A.2 F_U Master Equation (Complete â 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4} igl[\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}} igr]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
âÎ£Î»áµ¢Â·Uáµ¢Â·E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420):
Î»â=10â»Â¹â°, Î»â=10â»Â¹Â², Î»â=10â»Â¹Â¹, Î»â=10â»Â¹Â³ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} imesigl(1 + 10^{13}\,\Theta(ho_{SCm} - ho_c)igr) imesigl(1 + A_q \cos(\Delta\omega\, t)igr)$$

Symbol	Value	Description
Ï_c	10Â¹âµ kg/mÂ³	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
ÎÏ	2Ï/(434Â·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, â□¦)	Multi-scale field interactions
Buoyant	Î²_i Ã□ Ubi	Expanding nebulae, stellar winds
Superconductive	Um Ã□ (1+10Â¹Â³Â·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.